Decimos que una ecuación es reducible a homogénea o a variables separables si presenta la siguiente forma:

$$\frac{dy}{dx} = \frac{a_1 x + b_1 y + c_1}{a_2 x + b_2 y + c_2}$$

donde a_1; a_2; b_1; b_2; c_1; c_2 son constantes.

Ejemplos: $\frac{dy}{dx} = \frac{10x - 5y - 2}{4x - 2y + 7}$ $\qquad \frac{dx}{dy} = \frac{2 + 3x - 5y}{1 - x + 13y}$ $\qquad (2x - y + 4)dy + (x - 2y + 5)dx = 0$

Se pueden presentar dos casos:

El determinante $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$ es diferente de cero ($\Delta \neq 0$) $\longrightarrow$ Reducible a homogénea

El determinante $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$ es igual a cero ($\Delta = 0$) $\longrightarrow$ Reducible a variables separables

a) $\frac{dy}{dx} = \frac{x+y+4}{x-y-6}$ $\qquad a_1 = 1 \quad a_2 = 1 \quad b_1 = 1 \quad b_2 = -1$ $\qquad \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}$ $-1 - 1 = -2$

$\hookrightarrow$ Reducible a hom

$$\begin{cases} x + y = -4 \\ x - y = 6 \end{cases} \qquad x = 1 \quad y = -5$$

$u = x - 1$

$x = u + 1 \qquad dx = du$

$y = v - 5 \qquad dy = dv$

$v = y + 5$

$$\frac{dv}{du} = \frac{(u+\cancel{1}) + (v-\cancel{5}) + \cancel{4}}{(u+\cancel{1}) - (v-\cancel{5}) - \cancel{6}}$$

$$\frac{du}{dv} = \frac{u+v}{u-v}$$

$$\frac{du}{dv} = \frac{\frac{u+v}{v}}{\frac{u-v}{v}}$$

$t = \frac{u}{v} \rightarrow tv = u$

$$\frac{du}{dv} = \frac{\frac{u}{v} + 1}{\frac{u}{v} - 1}$$

$\frac{du}{dv} = t + \frac{dt}{dv} \cdot v$

$$t + \frac{dt}{dv} \cdot v = \frac{t+1}{t-1}$$

$$v \cdot \frac{dt}{dv} = \frac{t+1}{t-1} - t$$

$$v \cdot \frac{dt}{dv} = \frac{t+1-t^2+t}{t-1}$$

$$\frac{v\,dt}{dv} = \frac{-t^2+2t+1}{t-1}$$

$$\frac{dv}{v\,dt} = \frac{t-1}{-t^2+2t+1}$$

$$\int \frac{dv}{v} = -\int \frac{(t-1)dt}{+t^2-2t-1}$$

$u = t^2-2t-1$

$du = 2t-2\,dt$

$$\ln|v| = -\frac{1}{2}\ln|t^2-2t-1| + k$$

$$\ln|v^2(t^2-2t-1)| = 2k$$

$$\ln\left|v^2\left(\frac{u^2}{v^2}-2\frac{u}{v}-1\right)\right| = 2k$$

$$\ln|u^2-2uv-v^2| = 2k$$

$$\ln|(x-1)^2-2(x-1)(y+5)-(y+5)^2| = 2k$$

$$\ln|x^2-2x+1-2(xy+5x-y-5)-y^2+10y+25| = 2k$$

$$\ln|x^2-2x+1-2xy-10x+2y+10-y^2+10y+25| = 2k$$

$$\ln|x^2-y^2-2xy-12x+12y+36| = 2k$$

$$x^2-y^2-2xy-12x+12y+36 = c$$

$$\cancel{2x} - 2yy' - \cancel{2x} - 2xy' - 12 + 12y' = 0$$

$$-2yy' - 2xy' + 12y' = 12$$

$$y'(-2y-2x+12) = 12$$

$$y' = \frac{12}{(-2y-2x+12)}$$

$$\frac{dx}{\ } = x+y+4$$

$$\frac{\quad}{dy} = \frac{\quad}{x-y-6}$$